

The Contribution of Imports to Domestic Prices

Omar Barbiero

FRB Boston

Alvaro Silva

FRB Boston

Hillary Stein

FRB Boston

SED 2026

Disclaimer: *Views are those of the authors and not necessarily those of the Federal Reserve Bank of Boston, the Federal Reserve System, or any other institution.*

How do border price fluctuations impact consumer inflation?

- ▶ Tariffs in 2018–19 and 2025 passed through **fully** into U.S. **border** prices.

Amiti et. al (2019); Fajgelbaum et al. (2020); Cavallo et al. (2021); Amiti et. al (2026); Gopinath & Neiman (2026)

How do border price fluctuations impact consumer inflation?

- ▶ Tariffs in 2018–19 and 2025 passed through **fully** into U.S. **border** prices.

Amiti et. al (2019); Fajgelbaum et al. (2020); Cavallo et al. (2021); Amiti et. al (2026); Gopinath & Neiman (2026)

- ▶ Pass-through into domestic inflation still unclear
 - Micro studies find little; industry-level studies find a lot

How do border price fluctuations impact consumer inflation?

- ▶ Tariffs in 2018–19 and 2025 passed through **fully** into U.S. **border** prices.

Amiti et. al (2019); Fajgelbaum et al. (2020); Cavallo et al. (2021); Amiti et. al (2026); Gopinath & Neiman (2026)

- ▶ Pass-through into domestic inflation still unclear
 - Micro studies find little; industry-level studies find a lot

- ▶ Why? Downstream inflation depends on
 - The mix of final-goods vs. intermediate import exposure
 - How import cost shocks propagate through the domestic IO network
 - Market structure in the intermediate *and* retail sectors

How do border price fluctuations impact consumer inflation?

- ▶ Tariffs in 2018–19 and 2025 passed through **fully** into U.S. **border** prices.

Amiti et. al (2019); Fajgelbaum et al. (2020); Cavallo et al. (2021); Amiti et. al (2026); Gopinath & Neiman (2026)

- ▶ Pass-through into domestic inflation still unclear
 - Micro studies find little; industry-level studies find a lot
- ▶ Why? Downstream inflation depends on
 - The mix of final-goods vs. intermediate import exposure
 - How import cost shocks propagate through the domestic IO network
 - Market structure in the intermediate *and* retail sectors
- ▶ Different empirical designs estimate *different* empirical objects

How do border price fluctuations impact consumer inflation?

- ▶ Tariffs in 2018–19 and 2025 passed through **fully** into U.S. **border** prices.

Amiti et. al (2019); Fajgelbaum et al. (2020); Cavallo et al. (2021); Amiti et. al (2026); Gopinath & Neiman (2026)

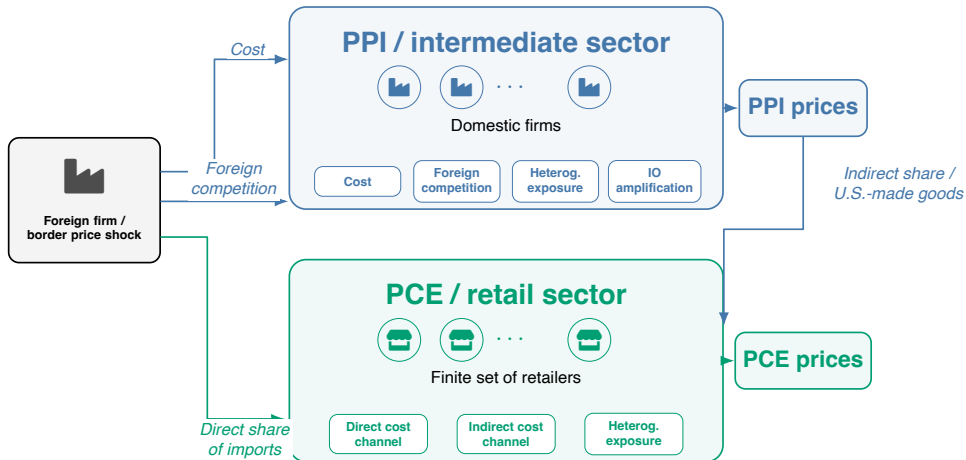
- ▶ Pass-through into domestic inflation still unclear
 - Micro studies find little; industry-level studies find a lot

- ▶ Why? Downstream inflation depends on
 - The mix of final-goods vs. intermediate import exposure
 - How import cost shocks propagate through the domestic IO network
 - Market structure in the intermediate *and* retail sectors

- ▶ Different empirical designs estimate *different* empirical objects

- ▶ This paper: provides a unified framework and take it to industry-level data

This paper: a price system response to a border shock



Preview: Empirical strategy

- ▶ Our theory gives us the empirical objects which we match to public data
 - BEA input-output accounts

Preview: Empirical strategy

- ▶ Our theory gives us the empirical objects which we match to public data
 - BEA input-output accounts
- ▶ We construct direct and indirect *model-predicted* tariff exposures

Preview: Empirical strategy

- ▶ Our theory gives us the empirical objects which we match to public data
 - BEA input-output accounts
- ▶ We construct direct and indirect *model-predicted* tariff exposures
- ▶ We compare *predicted* shocks with *observable* price changes using LP

Preview: Empirical strategy

- ▶ Our theory gives us the empirical objects which we match to public data
 - BEA input-output accounts
- ▶ We construct direct and indirect *model-predicted* tariff exposures
- ▶ We compare *predicted* shocks with *observable* price changes using LP
- ▶ We enrich the prediction in steps

Preview: Empirical strategy

- ▶ Our theory gives us the empirical objects which we match to public data
 - BEA input-output accounts
- ▶ We construct direct and indirect *model-predicted* tariff exposures
- ▶ We compare *predicted* shocks with *observable* price changes using LP
- ▶ We enrich the prediction in steps
 1. Pass-through in logs vs pass-through in levels

Preview: Empirical strategy

- ▶ Our theory gives us the empirical objects which we match to public data
 - BEA input-output accounts
- ▶ We construct direct and indirect *model-predicted* tariff exposures
- ▶ We compare *predicted* shocks with *observable* price changes using LP
- ▶ We enrich the prediction in steps
 1. Pass-through in logs vs pass-through in levels
 2. Oligopolistic competition

Preview: Empirical strategy

- ▶ Our theory gives us the empirical objects which we match to public data
 - BEA input-output accounts
- ▶ We construct direct and indirect *model-predicted* tariff exposures
- ▶ We compare *predicted* shocks with *observable* price changes using LP
- ▶ We enrich the prediction in steps
 1. Pass-through in logs vs pass-through in levels
 2. Oligopolistic competition
 3. Within-industry heterogeneity: are exposed firms the high-pass through ones?

Preview: Empirical strategy

- ▶ Our theory gives us the empirical objects which we match to public data
 - BEA input-output accounts
- ▶ We construct direct and indirect *model-predicted* tariff exposures
- ▶ We compare *predicted* shocks with *observable* price changes using LP
- ▶ We enrich the prediction in steps
 1. Pass-through in logs vs pass-through in levels
 2. Oligopolistic competition
 3. Within-industry heterogeneity: are exposed firms the high-pass through ones?
- ▶ Interpret which wedge can best match cross sectional inflation variation

Preview: Results

- ▶ 2018 tariff episode
 - baseline models understate tariff effect
 - room for a covariance wedge

Preview: Results

- ▶ 2018 tariff episode
 - baseline models understate tariff effect
 - room for a covariance wedge
- ▶ 2025 tariff episode
 - the baseline model seems to fit well
 - level specification somewhat better

Preview: Results

- ▶ 2018 tariff episode
 - baseline models understate tariff effect
 - room for a covariance wedge
- ▶ 2025 tariff episode
 - the baseline model seems to fit well
 - level specification somewhat better
- ▶ Reconcile with the fact that the 2018 episode was more targeted

Model Sketch

Framework to carry a border shock from PPI to PCE

- ▶ Shock observed at the product-country border

Framework to carry a border shock from PPI to PCE

- ▶ Shock observed at the product-country border
- ▶ Object of interest:

$$d \log p_F \xrightarrow{\text{other shocks fixed}} (d \log p^{\text{PPI}}; d \log p^{\text{PCE}})$$

Framework to carry a border shock from PPI to PCE

- ▶ Shock observed at the product-country border
- ▶ Object of interest:

$$d \log p_F \xrightarrow{\text{other shocks fixed}} (d \log p^{\text{PPI}}; d \log p^{\text{PCE}})$$

- ▶ Model ingredients:

Framework to carry a border shock from PPI to PCE

- ▶ Shock observed at the product-country border
- ▶ Object of interest:

$$d \log p_F \xrightarrow{\text{other shocks fixed}} (d \log p^{\text{PPI}}; d \log p^{\text{PCE}})$$

- ▶ Model ingredients:
 - task-assignment demand $\Rightarrow$ *pass-through in levels*
 - finite-firm Bertrand competition in both the producer and retail layer

Framework to carry a border shock from PPI to PCE

- ▶ Shock observed at the product-country border
- ▶ Object of interest:

$$d \log p_F \xrightarrow{\text{other shocks fixed}} (d \log p^{\text{PPI}}; d \log p^{\text{PCE}})$$

- ▶ Model ingredients:
 - task-assignment demand $\Rightarrow$ *pass-through in levels*
 - finite-firm Bertrand competition in both the producer and retail layer
 - domestic and imported inputs enter through an IO network
 - direct final-good import exposure in retail

Framework to carry a border shock from PPI to PCE

- ▶ Shock observed at the product-country border
- ▶ Object of interest:

$$d \log p_F \xrightarrow{\text{other shocks fixed}} (d \log p^{\text{PPI}}; d \log p^{\text{PCE}})$$

- ▶ Model ingredients:
 - task-assignment demand $\Rightarrow$ *pass-through in levels*
 - finite-firm Bertrand competition in both the producer and retail layer
 - domestic and imported inputs enter through an IO network
 - direct final-good import exposure in retail
 - aggregation from firms to observed PPI industries and PCE categories

▶ Model setup

Baseline model: full pass-through in levels vs. log

- Full pass-through in logs (standard): Multiplicative markups

$$p_i = \mu_i \cdot mc_i \quad \implies \quad d \log p_i = d \log mc_i$$

1% cost shock $\rightarrow$ 1% price increase

Baseline model: full pass-through in levels vs. log

- Full pass-through in logs (standard): Multiplicative markups

$$p_i = \mu_i \cdot mc_i \quad \implies \quad d \log p_i = d \log mc_i$$

1% cost shock $\rightarrow$ 1% price increase

- Full pass-through in levels: Additive markups

[Sangani, 2026]

$$p_i = \mu_i + mc_i \quad \implies \quad dp_i = dmc_i \quad \xrightarrow{\text{CRTS}} \quad d \log p_i = \frac{\text{Costs}_i}{\text{Sales}_i} d \log mc_i$$

1% cost shock $\rightarrow$ *cost-share* % price increase

► Pricing details

Baseline model: IO propagation in the intermediate market

- Pass-through in levels

$$d \log p_i^d = \Upsilon_i d \log mc_i$$

- Pass-through in *logs* is “nested” ($\Upsilon_i = 1$)

MAPPABLE DATA OBJECTS

Ω^d, Ω^*

input cost shares (dom./imp.)

Υ

variable cost / sales

Baseline model: IO propagation in the intermediate market

- Pass-through in levels

$$d \log p_i^d = \Upsilon_i d \log mc_i$$

- Pass-through in *logs* is “nested” ($\Upsilon_i = 1$)

- Domestic producers buy domestic and imported inputs under CRS:

$$d \log mc_i = \sum_j \Omega_{ij}^d d \log p_j^d + \sum_j \Omega_{ij}^* d \log p_j^*$$

MAPPABLE DATA OBJECTS

Ω^d, Ω^*

input cost shares (dom./imp.)

Υ

variable cost / sales

Baseline model: IO propagation in the intermediate market

- ▶ Chain cost shocks across industries $\Rightarrow$ vectorized PPI response:

$$d \log \mathbf{p}^d = (I - \Upsilon \Omega^d)^{-1} \Upsilon \Omega^* d \log \mathbf{p}^*$$

MAPPABLE DATA OBJECTS

Ω^d, Ω^*

input cost shares (dom./imp.)

Υ

variable cost / sales

Baseline model: IO propagation in the intermediate market

- ▶ Chain cost shocks across industries $\Rightarrow$ vectorized PPI response:

$$d \log \mathbf{p}^d = (I - \Upsilon \Omega^d)^{-1} \Upsilon \Omega^* d \log \mathbf{p}^*$$

- ▶ Take into account first + all downstream rounds of IO propagation:

$$(I - \Upsilon \Omega^d)^{-1} \Upsilon \Omega^* = \Upsilon \Omega^* + \Upsilon \Omega^d \Upsilon \Omega^* + \Upsilon (\Omega^d)^2 \Upsilon \Omega^* + \dots$$

MAPPABLE DATA OBJECTS

Ω^d, Ω^*

input cost shares (dom./imp.)

Υ

variable cost / sales

Baseline model: Retail markups layered on IO propagation

- Vectorized PPI response:

$$d \log \mathbf{p}^d = (I - \Upsilon \Omega^d)^{-1} \Upsilon \Omega^* d \log \mathbf{p}^*$$

MAPPABLE DATA OBJECTS

Ω^d, Ω^*

input cost shares (dom./imp.)

Υ, Υ^r

variable cost / sales

Θ^d, Θ^*

retail sourcing shares

Baseline model: Retail markups layered on IO propagation

- ▶ Vectorized PPI response:

$$d \log \mathbf{p}^d = (I - \Upsilon \Omega^d)^{-1} \Upsilon \Omega^* d \log \mathbf{p}^*$$

- ▶ Retailers buy domestic and imported final goods (no IO loop among retailers):

$$d \log \mathbf{p}^{\text{PCE}} = \Upsilon^r \left[\Theta^* + \Theta^d (I - \Upsilon \Omega^d)^{-1} \Upsilon \Omega^* \right] d \log \mathbf{p}^*$$

▶ Mapping details

MAPPABLE DATA OBJECTS

Ω^d, Ω^*

input cost shares (dom./imp.)

Υ, Υ^r

variable cost / sales

Θ^d, Θ^*

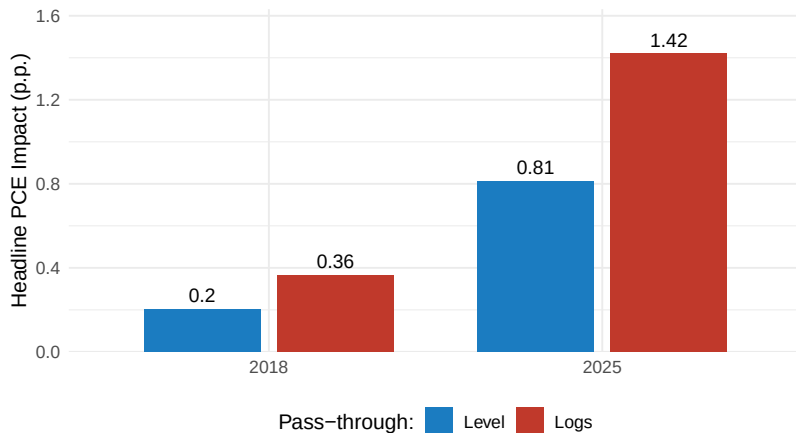
retail sourcing shares

Data and Shock

Data

- ▶ Several tables from the (BEA) Input-Output Accounts:
 - Use/Make Tables
 - Import Matrices
 - PCE Bridge
 - Commodity [= final good] level: $N = 402$
 - NIPA-code final expenditure category [= retail] level: $E = 212$
- ▶ Effective tariffs calculated at the product-country-month level
 - from public US Census data
 - HS-10-level tariffs are aggregated to BEA commodity categories
 - country-level import value shares from the year before the tariff episode

Estimated tariff impact of two tariff episodes: 2018-19 vs 2025

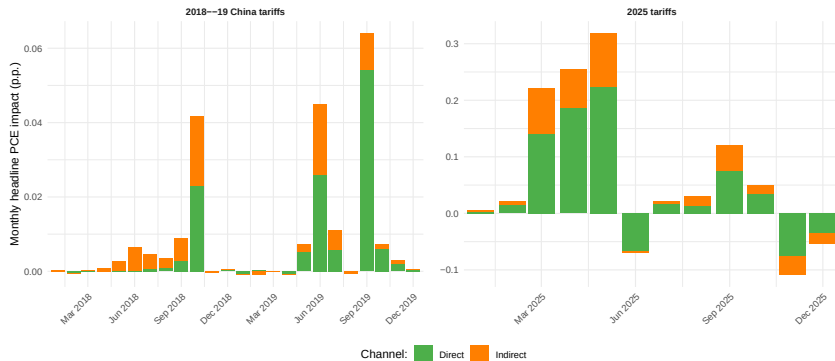


- ▶ 2018–19: concentrated on China (~90% of impact)
- ▶ 2025: broad-based across many source countries

Pass-through estimation under monopolistic competition

Pass-through estimation under monopolistic competition

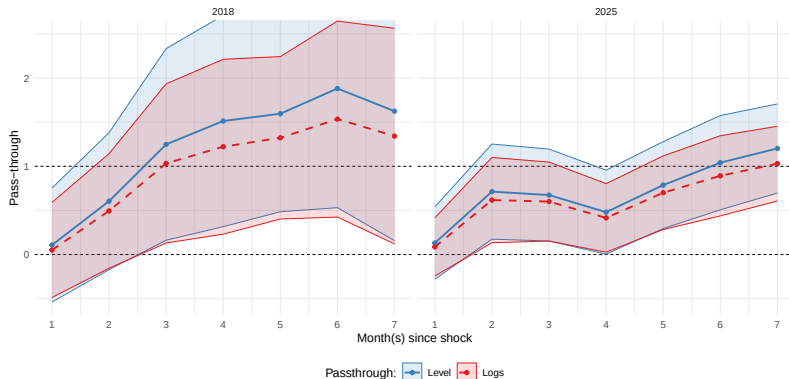
$$\Delta_h \log p_{i,t+h} = \alpha_i + \delta_t + \beta_h s_{i,t} + \sum_{k=-6, k \neq 0}^6 c_{h,k} s_{i,t-k} + \lambda_h \Delta \log w_{i,t-1} + \varepsilon_{i,t+h} \quad (1)$$



- $s_{i,t}$: model-implied tariff exposure of item i .
 - Feed $s_{i,t}$ into (1); compare β_h across episodes.
- $\beta_{\text{Long Horizon}} = 1 \Leftrightarrow$ model matches observed price change.

PPI: baseline model fits 2025, undershoots 2018

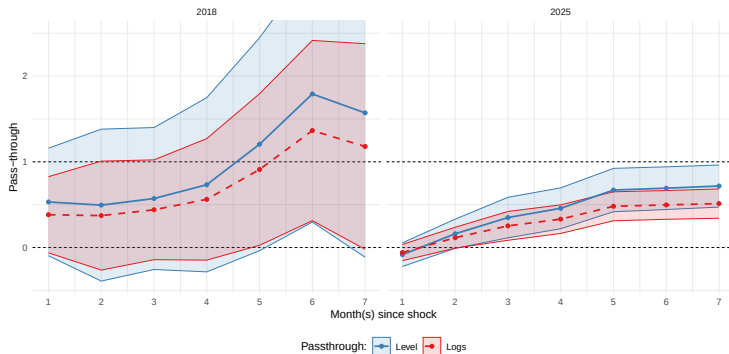
$$\Delta_h \log p_{i,t+h}^{\text{PPI}} = \alpha_i + \delta_t + \beta_h s_{i,t}^{\text{PPI}} + \sum_{k=-6, k \neq 0}^6 c_{h,k} s_{i,t-k}^{\text{PPI}} + \lambda_h \Delta \log w_{i,t-1} + \varepsilon_{i,t+h} \quad (2)$$



- **2018:** $\beta_h > 1$. Residual wedge?
- **2025:** long-horizon $\beta_h \approx 1$.

PCE: baseline model undershoots 2018-19, overshoots 2025

$$\Delta_h \log p_{i,t+h}^{\text{PCE}} = \alpha_i + \delta_t + \beta_h s_{i,t}^{\text{PCE}} + \sum_{k=-6, k \neq 0}^6 c_{h,k} s_{i,t-k}^{\text{PCE}} + \lambda_h \Delta \log w_{i,t-1} + \varepsilon_{i,t+h}$$



- ▶ 2018 overshoots $\beta_h=1$, 2025 undershoots.
- ▶ PCE is identified mostly from *direct* exposure $\implies$ pattern not driven by PPI.

Taking stock thus far

- ▶ Levels predicts less aggregate price response than logs
- ▶ Actual prices responded more than both would predict in 2018
 - under monopolistic competition!
- ▶ What else would increase or decrease the price response?

What explains the wedge?

Two wedges on top of the full pass-through baseline

$$d \log p_i^d = \frac{1}{\bar{\rho}_i^d + (1 - \bar{\rho}_i^d) s_{Fi}} \left[\bar{\rho}_i^d \Upsilon_i^d d \log \bar{m}c_i + (1 - \bar{\rho}_i^d) s_{Fi} d \log p_{Fi} + \Upsilon_i^d \text{Cov}_{\bar{\delta}_i}(\rho_i, d \log mc_i) \right]$$

Cost channel (baseline)

Monopolistic-competition benchmark. Explains some of the cross-sectional variation, both episodes.

Foreign competition (ρ)

Firms more sensitive to market price; high *foreign sales share* s_{Fi} implies markups increase with own-industry tariff.

Covariance

Are the exposed firms the high-pass-through ones? Within-industry heterogeneity in *who is hit*.

► Derivation

Foreign competition wedge can fit 2018 story, but not 2025

- ▶ Smaller $\rho \Rightarrow$ *larger* predicted impact.
 - Lower ρ means stronger foreign-competition pass-through.

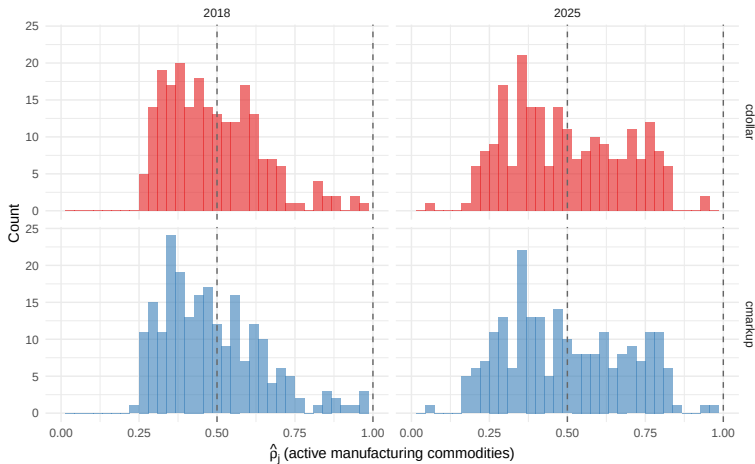
Foreign competition wedge can fit 2018 story, but not 2025

- ▶ Smaller $\rho \Rightarrow$ *larger* predicted impact.
 - Lower ρ means stronger foreign-competition pass-through.
- ▶ ρ_i is unobserved
 - a function of demand elasticities and market-share indices.

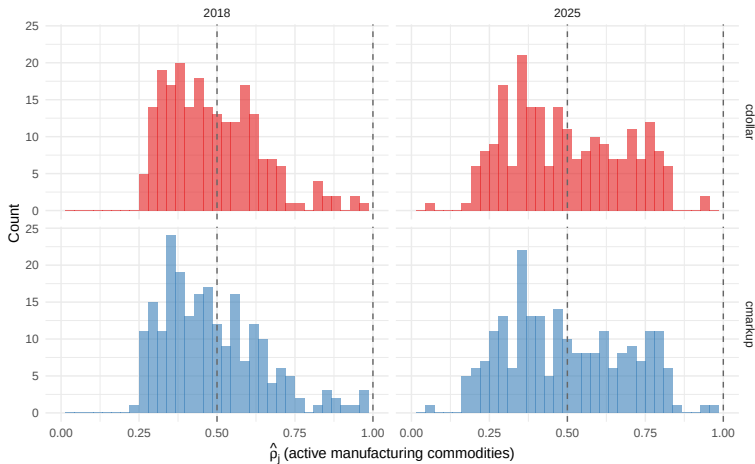
Foreign competition wedge can fit 2018 story, but not 2025

- ▶ Smaller $\rho \Rightarrow$ *larger* predicted impact.
 - Lower ρ means stronger foreign-competition pass-through.
- ▶ ρ_i is unobserved
 - a function of demand elasticities and market-share indices.
- ▶ GMM parametric identification from industry observables
 - HHI, markups, import share.

Foreign competition wedge can fit 2018 story, but not 2025



Foreign competition wedge can fit 2018 story, but not 2025



- ρ acts mostly as a residual shock absorber
 - smaller in 2018, nearer monopolistic competition in 2025.
 - ⇒ skeptical of foreign competition as a stand-alone story.

The additional story: shocks varied on *who* got them

- ▶ $\text{Cov}_{\tilde{\delta}_i}(\rho_i, d \log \mathbf{mc}_i)$ can plausibly vary across episodes
 - it captures the interaction between a *particular* shock and the firms exposed to it

The additional story: shocks varied on *who* got them

- ▶ $\text{Cov}_{\tilde{\delta}_i}(\rho_i, d \log \mathbf{mc}_i)$ can plausibly vary across episodes
 - it captures the interaction between a *particular* shock and the firms exposed to it
- ▶ **Work in Progress**: identify $\text{Cov}_{\tilde{\delta}_i}(\rho_i, d \log \mathbf{mc}_i)$ from industry-level observables.

The additional story: shocks varied on *who* got them

- ▶ $\text{Cov}_{\tilde{\delta}_i}(\boldsymbol{\rho}_i, d \log \mathbf{mc}_i)$ can plausibly vary across episodes
 - it captures the interaction between a *particular* shock and the firms exposed to it
- ▶ **Work in Progress**: identify $\text{Cov}_{\tilde{\delta}_i}(\boldsymbol{\rho}_i, d \log \mathbf{mc}_i)$ from industry-level observables.
- ▶ A possibility: impose regularity assumptions
 - Within-industry heterogeneity scales with import exposure

$$\text{Cov}_{\tilde{\delta}_i}(\boldsymbol{\rho}_i, \boldsymbol{\Omega}_{i,j}^*) = \mathbb{C}_{ij}^* = \gamma_i \boldsymbol{\Omega}_{ij}^*$$

- Identity domestic covariance matrix

$$\mathbb{C}^d = \mathbf{I}$$

The additional story: shocks varied on *who* got them

- ▶ $\text{Cov}_{\tilde{\delta}_i}(\boldsymbol{\rho}_i, d \log \mathbf{mc}_i)$ can plausibly vary across episodes
 - it captures the interaction between a *particular* shock and the firms exposed to it
- ▶ **Work in Progress**: identify $\text{Cov}_{\tilde{\delta}_i}(\boldsymbol{\rho}_i, d \log \mathbf{mc}_i)$ from industry-level observables.
- ▶ A possibility: impose regularity assumptions
 - Within-industry heterogeneity scales with import exposure

$$\text{Cov}_{\tilde{\delta}_i}(\boldsymbol{\rho}_i, \boldsymbol{\Omega}_{i,j}^*) = \mathbb{C}_{ij}^* = \gamma_i \boldsymbol{\Omega}_{ij}^*$$

- Identity domestic covariance matrix

$$\mathbb{C}^d = \mathbf{I}$$

- ▶ Two wedges, two loadings $\Rightarrow$ both separately identifiable:

The additional story: shocks varied on *who* got them

- ▶ $\text{Cov}_{\tilde{\delta}_i}(\rho_i, d \log \mathbf{mc}_i)$ can plausibly vary across episodes
 - it captures the interaction between a *particular* shock and the firms exposed to it
- ▶ **Work in Progress**: identify $\text{Cov}_{\tilde{\delta}_i}(\rho_i, d \log \mathbf{mc}_i)$ from industry-level observables.
- ▶ A possibility: impose regularity assumptions
 - Within-industry heterogeneity scales with import exposure

$$\text{Cov}_{\tilde{\delta}_i}(\rho_i, \Omega_{i,j}^*) = \mathbb{C}_{ij}^* = \gamma_i \Omega_{ij}^*$$

- Identity domestic covariance matrix

$$\mathbb{C}^d = I$$

- ▶ Two wedges, two loadings $\Rightarrow$ both separately identifiable:
 - γ_i loads on Ω^* — re-weights the firms actually exposed to imports.

The additional story: shocks varied on *who* got them

- ▶ $\text{Cov}_{\tilde{\delta}_i}(\rho_i, d \log \mathbf{mc}_i)$ can plausibly vary across episodes
 - it captures the interaction between a *particular* shock and the firms exposed to it
- ▶ **Work in Progress**: identify $\text{Cov}_{\tilde{\delta}_i}(\rho_i, d \log \mathbf{mc}_i)$ from industry-level observables.
- ▶ A possibility: impose regularity assumptions
 - Within-industry heterogeneity scales with import exposure

$$\text{Cov}_{\tilde{\delta}_i}(\rho_i, \Omega_{i,j}^*) = \mathbb{C}_{ij}^* = \gamma_i \Omega_{ij}^*$$

- Identity domestic covariance matrix

$$\mathbb{C}^d = I$$

- ▶ Two wedges, two loadings $\Rightarrow$ both separately identifiable:
 - γ_i loads on Ω^* — re-weights the firms actually exposed to imports.
 - $(1 - \rho_i)$ loads on foreign-competition share $\mathbf{S}_F$.

The additional story: shocks varied on *who* got them

- ▶ $\text{Cov}_{\tilde{\delta}_i}(\rho_i, d \log \mathbf{mc}_i)$ can plausibly vary across episodes
 - it captures the interaction between a *particular* shock and the firms exposed to it
- ▶ **Work in Progress**: identify $\text{Cov}_{\tilde{\delta}_i}(\rho_i, d \log \mathbf{mc}_i)$ from industry-level observables.
- ▶ A possibility: impose regularity assumptions
 - Within-industry heterogeneity scales with import exposure

$$\text{Cov}_{\tilde{\delta}_i}(\rho_i, \Omega_{i,j}^*) = \mathbb{C}_{ij}^* = \gamma_i \Omega_{ij}^*$$

- Identity domestic covariance matrix

$$\mathbb{C}^d = I$$

- ▶ Two wedges, two loadings $\Rightarrow$ both separately identifiable:
 - γ_i loads on Ω^* — re-weights the firms actually exposed to imports.
 - $(1 - \rho_i)$ loads on foreign-competition share $\mathbf{S}_F$.
 - PCE: retail has no foreign-competition term ($\rho^r = 1$)
 - $\Rightarrow \gamma^r$ identified *directly* via GMM from direct-exposure variation.

Takeaways

- ▶ A single framework nests
 - direct exposure
 - IO propagation
 - levels-vs-logs pass-through
 - oligopolistic competition
- ▶ Baseline full pass-through of cost channel explains
 - most of PPI and PCE cross-sectional variation in *both* episodes.
 - But 2018 inflation responded more than what the model predicts
 - vice versa for 2025
- ▶ Neither levels-vs-logs nor the foreign-competition stably explains both episodes.
- ▶ Residual story seems to be about *who* is exposed within an industry
 - Work in progress!

Thank you!

asilvub.github.io

Appendix

Where this sits: pass-through × networks × imperfect competition

1. **Tariff pass-through.** Amiti, Redding & Weinstein (2019); Fajgelbaum et al. (2020); Cavallo, Llamas & Vazquez (2025); Flaaen & Pierce (2024); Minton & Somale (2025).
 - We clarify *which* downstream object each design is identifying.
2. **Production networks & inflation.** Baqaee & Farhi (2019); La'O & Tahbaz-Salehi (2022); Baqaee & Rubbo (2023).
 - We map border shocks through IO while keeping imperfect competition and firm-level heterogeneity.
3. **Markups & imperfect competition.** Atkeson & Burstein (2008); Amiti, Itskhoki & Konings (2019); Sangani (2026).
 - We embed pass-through *in levels* and oligopolistic competition with foreign firms into an IO + retail model.

From firm pricing to industry PPI: aggregation under oligopoly

N domestic firms f compete with each other and with one foreign firm F . Under pass-through in levels, $dp_{fi} \equiv d\mu_{fi} + dmc_{fi}$.

1. Firm price.

$$d \log p_{fi} = \rho_{fi} \Upsilon_{fi} d \log mc_{fi} + (1 - \rho_{fi}) \frac{p_{-f,i}}{p_{fi}} d \log p_{-f,i}$$

2. Industry aggregator. Market clearing delivers share-weighted prices:

$$d \log p_i = \sum_f s_{fi}^d d \log p_{fi}^d + s_{Fi} d \log p_i^*$$

3. Domestic PPI. Substitute and collect:

$$d \log p_i^d = \frac{1}{\bar{\rho}_i^d + (1 - \bar{\rho}_i^d) s_{Fi}} \left[\bar{\rho}_i^d \Upsilon_i^d d \log \bar{mc}_i + \Upsilon_i^d \text{Cov}_{\tilde{\delta}_i}(\rho_i, d \log mc_i) + \frac{(1 - \bar{\rho}_i^d) s_{Fi}}{\Upsilon_i^d} d \log p_{Fi} \right]$$

$$\rho_{fi} = \frac{(1 - \pi_{fi})^2}{1 - \pi_{fi} + \pi_{fi}^2}, \text{ firm cost pass-through}$$

$$\bar{\rho}_i^d = \sum_{f \in \mathcal{D}_i} \tilde{\delta}_{fi} \rho_{fi}, \text{ cost-share-weighted avg.}$$

$$\tilde{\delta}_{fi} = \frac{s_{fi}^d (1 - \alpha_{fi})}{\sum_g s_{gi}^d (1 - \alpha_{gi})}, \text{ firm variable-cost share}$$

$$d \log \bar{mc}_i = \sum_f \tilde{\delta}_{fi} d \log mc_{fi}, \text{ avg. cost shock}$$

$$\Upsilon_i^d = VC_i / R_i, \text{ industry cost-over-sales ratio}$$

$$s_{Fi} \text{ foreign-sales share in industry } i$$

$$\text{Cov}_{\tilde{\delta}_i}(\rho_i, d \log mc_i) = \sum_f \tilde{\delta}_{fi} (\rho_{fi} - \bar{\rho}_i^d) (d \log mc_{fi} - d \log \bar{mc}_i), \text{ firm-level covariance}$$

Full system with the covariance wedge

Under the regularity assumption $\text{Cov}_{\delta_i}(\rho_i, d \log mc_i) = \gamma_i \Omega_{ij}^*$, the PPI system becomes

$$d \log \mathbf{p}^d = \Xi^{-1}[(\rho^d + \gamma) \Upsilon^d \Omega^* + (I - \rho^d) \mathbf{S}_F] d \log \mathbf{p}^*$$

Ξ : block collecting domestic-propagation terms

ρ^d : cost-weighted industry pass-through

γ : diagonal covariance wedge

$\mathbf{S}_F$: diagonal of foreign-sales shares

Model details: assignment primitives

- ▶ Industry $i \in \mathcal{I}$ has domestic firms $\mathcal{D}_i$ and one foreign firm F_i .
- ▶ A continuum of tasks is assigned to discrete firms; firm f receives task share π_{fi} .
- ▶ Final-demand categories $e \in \mathcal{E}$ use the same assignment structure, but all firms are domestic retailers.

$$p_i^{\text{eff}} = \min_{\pi_i \in \Delta_i} \left\{ b_i + \sum_{f \in \mathcal{N}_i} \pi_{fi} (p_{fi} - z_{fi}) + \sigma_i \sum_{f \in \mathcal{N}_i} \pi_{fi} \log \pi_{fi} \right\}$$
$$\pi_{fi} = \frac{\exp((z_{fi} - p_{fi})/\sigma_i)}{\sum_{g \in \mathcal{N}_i} \exp((z_{gi} - p_{gi})/\sigma_i)}, \quad q_{fi} = \pi_{fi} q_i, \quad R_{fi} = p_{fi} \pi_{fi} q_i.$$

Model details: pricing and pass-through

- ▶ Domestic producers and retailers choose prices under Bertrand competition.
- ▶ The logit assignment block delivers an additive markup:

$$\Pi_{fi} = \max_{p_{fi}} (p_{fi} - mc_{fi}) \pi_{fi} q_i \implies p_{fi} = mc_{fi} + \frac{\sigma_i}{1 - \pi_{fi}}.$$

Levels.

$$dp_{fi} = (1 - \pi_{fi}) dmc_{fi} + \pi_{fi} dp_{-f,i}$$

$1 - \alpha_{fi} = mc_{fi}/p_{fi}$ is the cost share in the price. Even with complete pass-through in levels, percentage pass-through is attenuated by the cost share.

Percentage changes.

$$d \log p_{fi} = (1 - \alpha_{fi})(1 - \pi_{fi}) d \log mc_{fi} + \pi_{fi} \frac{p_{-f,i}}{p_{fi}} d \log p_{-f,i}$$

[◀ back](#)[▶ mapping](#)

Model details: observed indices and data mapping

- Observed PPI is a domestic-sales weighted index; marginal costs aggregate through observed input shares:

$$d \log p_i^d = \sum_{f \in \mathcal{D}_i} s_{fi}^d d \log p_{fi}, \quad d \log mc_i^d = \sum_j \Omega_{ij}^d d \log p_j^d + \sum_j \Omega_{ij}^* d \log p_j^*.$$

$$d \log \mathbf{p}^d = (\mathbf{I} - \mathbf{\Upsilon}^d \mathbf{\Omega}^d)^{-1} \mathbf{\Upsilon}^d \mathbf{\Omega}^* d \log \mathbf{p}^*.$$

- Retailer marginal costs combine domestic sourcing and direct imported final goods:

$$d \log \mathbf{p}^{\text{PCE}} = \mathbf{\Upsilon}^r \left[\mathbf{\Theta}^d (\mathbf{I} - \mathbf{\Upsilon}^d \mathbf{\Omega}^d)^{-1} \mathbf{\Upsilon}^d \mathbf{\Omega}^* + \mathbf{\Theta}^* \right] d \log \mathbf{p}^*.$$

$\mathbf{\Omega}$ maps imported-input shocks into producer marginal costs; $\mathbf{\Theta}$ maps producer prices and imported finished goods into PCE categories.